

Graphical Abstract

RaceSegNet: A Fairness-Driven Expert System for Demographic-Aware Segmentation and Classification via Multi-Task Learning

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Highlights

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- Creation of a large-scale, pixel-level annotated dataset with automated masks and metadata, reducing annotation costs by 90
- Development of a multi-task learning architecture integrating RACR and RASE modules for precise color-coded masks.
- Introduction of RaceSegNet for accurate racial region segmentation and improved cross-dataset generalization.
- Benchmarking demonstrates RaceSegNet outperforms SOTA models with superior fairness and generalization.

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Abstract

To address the problem of demographic bias in pixel-level race classification base face segmentation, we propose a demographic fairness aware framework that accomplishes dual challenges of racial annotations in existing datasets and architectural limitations in preserving demographic fairness. Traditional approaches suffer from coarse segmentation and racial accuracy gaps due to their failure to discriminate diverse demographic features from facial structures. We introduce RaceSegNet, a parallel dual-branch network with pyramid feature extractors, optimized for fair demographic aware segmentation. The segmentation branch employs Progressive Refinement Heads (PRH) using iterative enhanced approaches for fine-grained racial regions (e.g., skin, nose, eyes), while Boundary Enhancement Modules (BEM) utilize edge uncertainties through learned spatial attention the proposed model structure can unified multi tasking race classification and face segmentation. pixel level race labeling dataset limitations and not open source available, to overcome this we design in our preprocessing pipeline that fuses CelebMask-HQ facial components with racial labels from two race balanced datasets, FairFace and UTKFace, using Race-Identity Aware Channel Refinement (RACR) to align features with demographic attributes and Race-Aware Spatial Enhancement (RASE) that introduces label to pixel level masks. This preprocessing pipeline automatically generates 120,000+ color coded demographically balanced masks without manual annotation ensuring privacy compliance. RaceSegNet achieves state-of-the-art performance, with a face IoU of 0.921 on FairFace and 0.890 on UTKFace, outperforming baselines on the minority class. Cross-dataset validation demonstrates the proposed model’s robustness while maintaining efficiency (2.84M parameters). This work establishes

a foundation for bias resilient applications in surveillance, healthcare, and human computer interaction where demographic fairness is essential.

Keywords: Face Segmentation, Race Classification, Demographic-Aware Learning, Race-Aware Attention Mechanisms, Progressive Refinement Network, Adaptive Feature Fusion

1. Introduction

Facial analysis is a foundational pillar of computer vision, enabling the extraction of soft biometric cues such as age, gender, and race [1, 2, 3]. Facial features encode rich demographic information through anthropometric measurements [4], with race being a critical factor across technological, medical, and social domains [5]. Traditional race classification, however, often assigns a single global label to an entire image. This approach is inherently limited: it collapses all facial pixels and background information into one category, fails to capture local feature variations, and is susceptible to bias from non-facial elements like lighting, occlusions, or scene context [6, 7]. Consequently, models may be influenced by the background elements, leading to reduced accuracy and fairness, particularly in demographically diverse or multi-ethnic settings [8, 9, 10].

To overcome these limitations, we propose a unified framework that combines face segmentation and race classification to produce a shared representation that localizes the face and predicts its race simultaneously. This integrated approach ensures the model’s focus is directed to race-discriminative facial regions within a precise region-of-interest (ROI), thereby reducing background bias and improving the classification task. By performing segmentation and classification concurrently through a multi-task architecture, the model learns to identify and leverage the most relevant facial features such as skin texture, eye shape, and lip morphology, while actively ignoring deceptive background information. This leads to more robust, accurate, and fair predictions, which is vital for sensitive applications in surveillance, forensic analysis, healthcare, and human-computer interaction [11, 12, 13].

Most existing methods treat race prediction as a pure classification problem, lacking the spatial granularity to explain which facial regions contribute to the decision. This obscures interpretability and hinders bias correction. Semantic, pixel-level race-aware segmentation addresses this by assigning demographic labels to each facial pixel, preserving spatial structure and en-

abling fine-grained analysis. Yet, this task remains largely unexplored due to the absence of pixel-annotated datasets; resources like MORPH, UTKFace, and FairFace [14, 15, 18] only provide image-level labels. Furthermore, ethical concerns regarding bias and the underrepresentation of minority groups in training data persist, often leading to uneven performance across demographics [16].

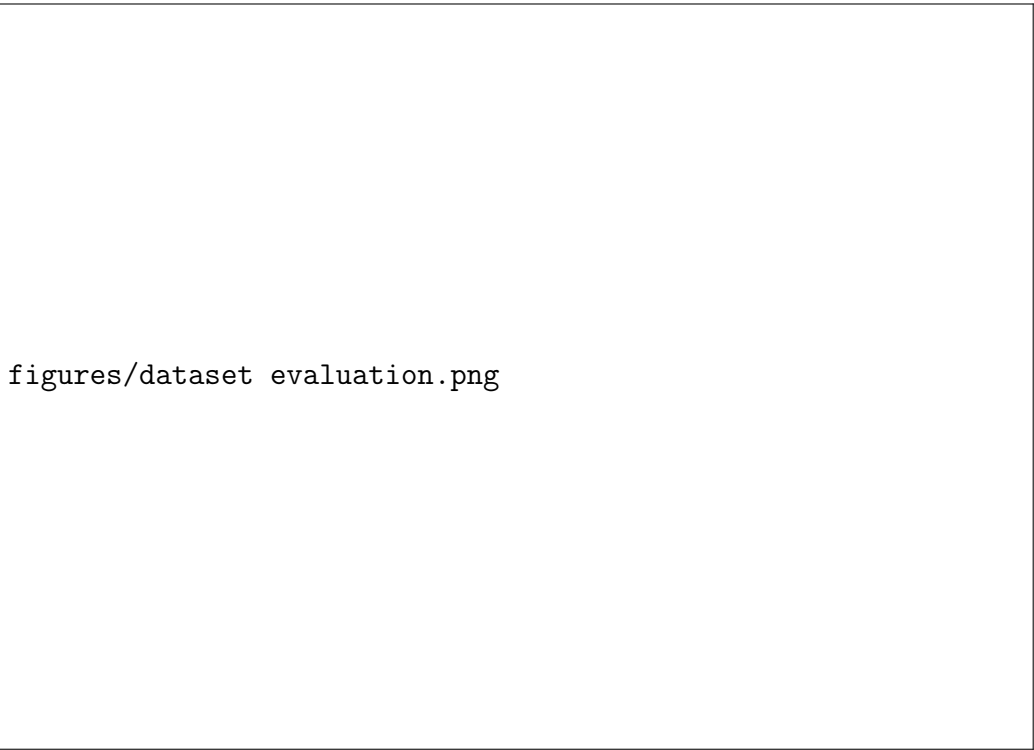
In this work, we address this critical gap through a comprehensive two-part approach. First, we construct a novel pixel-level annotated dataset by fusing detailed facial component masks from CelebAMask-HQ [17] with demographic labels from FairFace and UTKFace, resulting in a collection of race-aware, color-coded segmentation masks. Second, we propose RaceSegNet, a parallel dual-branch network built upon a shared pyramid feature extractor. The architecture is specifically designed for the joint task of demographic-aware face segmentation and race classification. It incorporates novel Race-Identity Aware Channel Refinement (RACR) and Race-Aware Spatial Enhancement (RASE) modules to adaptively highlight discriminative features, alongside Progressive Refinement and Boundary Enhancement Modules to ensure fine-grained segmentation and robust generalization across diverse demographics. This unified learning paradigm represents a significant advancement toward precise, interpretable, and equitable facial analysis systems. Extensive experiments on FairFace and UTKFace datasets demonstrate that RaceSegNet outperforms current state-of-the-art segmentation models in both performance and generalization.

2. Related Work

2.1. Race classification with various datasets

Research on race classification using facial images has emphasized the evolution of methods alongside challenges in dataset diversity and model accuracy. Early work focused on improving performance, defining racial categories, and refining feature extraction. Foundational surveys [1, 4] provided comprehensive reviews of race recognition, ethnicity prediction, and the application of deep learning to facial analysis, systematically exploring dimensions from feature representation to real-world applications.

A key contribution was the development of the MORPH-II dataset, a large-scale collection of 55,000 facial images [3]. In one study, a feature extraction technique based on biologically inspired features (BIF) achieved high accuracy for Black (98.3%) and White (97.1%) classes, but performance was



figures/dataset evaluation.png

Figure 1: Comparison of Key Quality Metrics Across Publicly Available Facial Recognition Datasets. This radar chart compares datasets based on class balance, diversity score, annotation accuracy, resolution consistency, label completeness, noise level, bias score, and privacy compliance.

lower for Hispanic (74.2%) and dramatically poor for Indian (6.9%) faces. This disparity highlights the difficulty in creating universally accurate models. Other datasets like UTKFace, FairFace, LFW, and FOTW provide more racial classes to help overcome such biases. To compare the quality of these datasets, Figure 1 [14, 18] presents key metrics, including class balance, diversity score, and annotation accuracy in a radar chart. This analysis shows that while datasets like FairFace and UTKFace offer good diversity, others struggle with annotation accuracy and noise, underscoring the persistent issue of dataset bias.

2.2. Race Classification

Pixel-level race classification and face segmentation overcome this by segmenting the facial region and assigning a race label to each pixel, enabling

the model to focus on key race-discriminative facial features within the specified ROI this provides a more accurate and interpretable race classification, particularly in applications like surveillance, healthcare, and crowd analysis, where precise and fair race identification is critical. Unlike traditional datasets such as UTKFace and FairFace [14, 15], which focuses on global race labels, pixel-level segmentation ensures the model can handle racial features within specified boundaries, improving multi ethnic classification and reducing bias in demographic analysis. This approach ultimately leads to better fairness and more robust race classification by focusing on the facial region that truly informs racial identity.

In contrast to pixel-aware segmentation approaches, earlier methods often suffered limitations due to their reliance on holistic facial features. For example, Gudi et al. [19] employed a CNN architecture with convolutional layers, local context normalization (LCN), and max-pooling operations to extract race-specific characteristics. Although this method achieved strong performance for majority groups such as Caucasian and East Asian individuals, the model performed poorly on underrepresented classes, with the mean average precision (mAP) dropping to 61.52%. These results highlight the challenges of learning robust race representations in the presence of severe class imbalance.

Similarly, Anwar [20] proposed a large-scale race classification system using handcrafted feature extraction combined with an SVM classifier. While computationally efficient, the absence of segmentation mechanisms limited the framework’s ability to capture fine-grained racial cues in localized regions of the face. Without the pixel-level understanding of structural boundaries, the model struggled to differentiate subtle variations across multi-ethnic groups. Overall, these earlier approaches underline the importance of segmentation-driven, pixel-centric strategies for achieving fair, balanced, and high-resolution race classification.

Al Humadin et al. [21] addressed the lack of Arab-focused datasets in race and ethnicity research by constructing a dedicated Arab facial image dataset labeled with three sub-ethnic groups: Gulf Cooperation Council (GCC), Levant, and Egyptian. Their work concentrated solely on Arabic ethnicity classification, employing both supervised and unsupervised deep learning approaches. For supervised learning, they utilized a pre-trained CNN model to classify the three sub-ethnic labels, achieving a best accuracy of 56.97% when trained on the full dataset. They further introduced deep clustering as an unsupervised alternative marking the first use of such methods in ethnicity

classification obtaining results ranging from 32% to 59% accuracy depending on the dataset and evaluation measure. Their findings highlight the difficulty of generalizing Arabic sub-ethnicity classification across diverse datasets and the need for more comprehensive representation of Arab populations in facial recognition research.

Liu [22] introduced one of the earliest large-scale frameworks for multi-attribute facial segmentation using the CelebA dataset, enabling the model to identify and segment multiple facial components simultaneously. Their face attributes prediction approach used a deep convolutional architecture to extract local face regions such as eyes, nose, mouth, and hair, demonstrating how fine-grained segmentation enhances downstream tasks like identity verification and attribute prediction. Although, their work established the foundation for region-level parsing, showing that ROI segmentation provides richer, more interpretable features compared to global face analysis.

The [23] address the key limitation of existing face parsing methods, which struggle to accurately segment long-tail facial attributes such as hats, earrings, and eyeglasses due to the dominance of common facial regions during CNN training. To solve this problem, they propose SegFace, a lightweight transformer-based model that uses class-specific tokens to independently model each facial component. This design significantly improves segmentation quality for rare classes while maintaining strong performance on common attributes. Their approach achieves state-of-the-art results on CelebAMask-HQ and LaPa datasets and delivers real-time processing speeds, marking a notable advancement in fine-grained, multi-attribute face segmentation.

Benini [24] introduced an early and influential framework for semantic face segmentation aimed at improving downstream face analysis tasks. Their approach segmented each facial image into six semantic regions—hair, skin, nose, eyes, mouth, and background—using a Random Forest classifier trained on their manually annotated FASSEG dataset. By combining location, shape, and color features, their method provided reliable pixel-wise segmentation and subsequently enhanced performance in head pose estimation, gender recognition, and expression classification. Evaluated across multiple standard datasets, the framework consistently outperformed existing methods, demonstrating that accurate facial segmentation can significantly strengthen face analysis pipelines.

study extended segmentation-guided analysis by proposing a framework for segmentation and classification task via automatic gender classification using facial part segmentation as a feature extraction stage [25]. The authors

trained a CRF-based semantic segmentation model to divide facial images into six key regions hair, skin, eyes, nose, mouth, and background—before generating probability maps for each component. These maps were then used to train a Random Decision Forest classifier capable of distinguishing between male and female faces. The integration of segmentation improved discrimination by isolating gender-informative regions, leading to state-of-the-art results on datasets such as Adience, LFW, FERET, and FEI. Despite this success, the method remains constrained by its dependency on accurate segmentation, and CRF-based models are less effective when faces exhibit occlusions, pose variations, or low-quality visual conditions.

A more recent direction in segmentation-classification research focuses on face forgery detection, where the Lin [26] proposed a meta-learning framework capable of segmenting manipulated regions and classifying forged images using only a few training samples. Unlike traditional forgery detectors that depend on large paired datasets of real and artificially generated images, their method uses meta-learning to rapidly adapt to new and unseen forgery types. The model learns segmentation and classification jointly by updating its weights based on the statistical characteristics of a small number of input fake images. This approach achieved substantial improvements in IoU and AUC across multiple forgery types, outperforming existing state-of-the-art systems. However, its performance still depends on the availability and representativeness of few-shot examples, and extreme or highly novel manipulation techniques may still challenge its generalization.

The Mahbub [27] introduces the SPLITFACE framework a segmentation-driven approach to facial attribute classification by explicitly handling partially occluded faces a limitation in most existing full-face-dependent models. Instead of relying solely on a complete facial image, the method segments the face into multiple localized regions and extracts both segment-level and full-face representations. Its deep convolutional architecture learns which attributes are best predicted from which facial segments, enabling a committee-machine strategy that fuses local and global predictions for improved attribute classification. This segmentation-guided classification design is particularly effective when portions of the face are hidden, as the model can still accurately classify attributes that are visible in the remaining segments. Evaluated on CelebA and LFWA, including custom versions with synthetic occlusions, SPLITFACE consistently outperforms other attribute detection models, especially under partial-face conditions highlighting the advantage of segment-level feature extraction for robust and reliable attribute

classification.

3. Methodology

This study proposes a unified deep learning framework that jointly learns face segmentation and race classification through a shared feature representation. The core hypothesis is that by learning to localize facial structures via segmentation, the model is compelled to focus on anthropologically relevant regions (e.g., skin texture, facial bone structure, eye shape), thereby reducing spurious correlations from the background and improving the robustness and fairness of race classification. The architecture is designed so that both tasks inform and refine a common feature embedding, which simultaneously encodes "where" the face is and "what" its racial characteristics are.

3.1. Dataset Transformation for Joint Learning

To enable joint segmentation and classification, facial images are transformed into race-aware segmentation masks. An automated pipeline processes images from the CelebMask-HQ dataset: it extracts facial ROIs, assigns a race label, and generates a color-coded segmentation mask where each color corresponds to a specific race class.

Formally, let an input image be $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$. It is resized to $\mathbf{I}' \in \mathbb{R}^{256 \times 256 \times 3}$. A corresponding mask $\mathbf{M} \in \mathbb{R}^{256 \times 256}$ is created, where its pixel value indexes race classes. For visualization, each race class is given an RGB color. The relationship of race, index and RGB color is given in Table 1. Pixels not belonging to the facial ROI are assigned a background label -1. This paired data $(\mathbf{I}', \mathbf{M})$ forms the basis for multi-task learning, where the model must reconstruct the race-indexed mask while predicting the overall race label.

3.2. The Proposed RaceSegNet

RaceSegNet is a unified multi-task convolutional framework designed to jointly perform facial region segmentation and human race classification within a single end-to-end trainable architecture. The key motivation is to exploit the strong spatial priors learned during facial segmentation to guide race classification toward ethnically discriminative facial regions, while preserving computational efficiency.

As illustrated in 2, the network consists of three major components:

Table 1: Unified race mask color mapping showing the RGB values and race categories for both the FairFace (7 classes) and UTKFace (5 classes) datasets.

Dataset	Race ID	Race Name	RGB Color (R,G,B)
FairFace	0	White	(0, 215, 255)
FairFace	1	Black	(51, 153, 255)
FairFace	2	Latino_Hispanic	(19, 69, 139)
FairFace	3	East Asian	(250, 250, 255)
FairFace	4	Southeast Asian	(152, 251, 152)
FairFace	5	Indian	(238, 130, 238)
FairFace	6	Middle Eastern	(203, 192, 255)
UTKFace	0	White	(255, 250, 250)
UTKFace	1	Black	(139, 69, 19)
UTKFace	2	Asian	(255, 215, 0)
UTKFace	3	Indian	(255, 153, 51)
UTKFace	4	Others	(152, 251, 152)

- A shared multi-scale feature encoder based on a dual-branch design,
- A segmentation branch with boundary-aware progressive refinement,
- A race classification branch equipped with race-aware attention mechanisms.

The shared encoder extracts rich hierarchical representations that are simultaneously optimized for both pixel-level and image-level tasks.

3.2.1. Shared Multi-Scale Feature Encoder

Given an RGB facial image $\mathbf{X} \in \mathbb{R}^{3 \times H \times W}$, the network first applies a shallow convolutional layer to obtain low-level features,

$$\mathbf{F}_0 = \text{BN}(\text{ReLU}(\text{Conv}_{3 \times 3}(\mathbf{X}))), \quad \mathbf{F}_0 \in \mathbb{R}^{32 \times H \times W}. \quad (1)$$

To capture both global semantic context and fine-grained facial structures, $\mathbf{F}_0$ is processed by a Dual-Branch Feature Extractor (DBFE):

- a stride-4 branch for coarse, high-level semantics,
- a stride-2 branch for spatially detailed representations.

figures/main model.png

Figure 2: Comprehensive overview of the proposed network modules. (a) Dual Branch Feature Extractor processes input through parallel stride-4 and stride-2 paths with concatenation fusion. (b) Pyramid Feature Extractor generates multi-scale representations. (c) Boundary Enhancement Module employs X and Y Sobel operators for edge-aware feature extraction (output: [B,2,H,W]). (d) Progressive Refine Head performs coarse and ultra-fine segmentation through CS+BE and FS+BE stages. (e) Spatial Attention Module reduces channels from C0=288 to C'=C/8 using global average and max pooling. Left: Dataset preprocessing showing color-coded facial masks, class labels, and ROI annotations for multi-task learning.

After multi-stage convolution and resolution alignment, the outputs of both branches are concatenated with $\mathbf{F}_0$:

$$\mathbf{F}_{cn} = \text{Concat}(\mathbf{F}_0, \text{Up}(\mathbf{F}_1), \text{Up}(\mathbf{F}_2)) \in \mathbb{R}^{288 \times H \times W}. \quad (2)$$

This fused representation serves as the shared feature backbone for subsequent task-specific heads.

3.2.2. Pyramid Feature Extraction and Boundary Enhancement for Segmentation

For the segmentation task, the fused feature map $\mathbf{F}_{cn}$ is first processed by a Pyramid Feature Extractor (PFE) to model scale-invariant facial structures using parallel convolutions with different receptive fields (1×1 , 3×3 , 5×5 , 7×7). The resulting multi-scale features are adaptively fused via learnable soft weights.

To further improve boundary localization, a Boundary Enhancement Module (BEM) explicitly incorporates gradient cues extracted using Sobel operators. These edge features are embedded and fused with the pyramid-enhanced features, producing a boundary-aware representation:

$$\mathbf{F}_{BE} \in \mathbb{R}^{128 \times H \times W}. \quad (3)$$

Finally, a Progressive Refinement Head (PRH) performs coarse-to-fine segmentation through three stages (coarse, fine, ultra-fine), allowing gradual correction of spatial inaccuracies and precise delineation of facial regions.

3.2.3. Race-Aware Attention for Classification

While the segmentation branch focuses on pixel-level structure, race classification relies on identifying globally discriminative facial cues such as skin tone distribution, facial geometry, and region-specific textures. To this end, RaceSegNet introduces a Race-Aware Attention mechanism applied to the shared feature representation.

Race-Identity Aware Channel Refinement (RACR) RACR emphasizes channels that are most informative for race discrimination. Given an input feature map $\mathbf{F} \in \mathbb{R}^{C \times H \times W}$, a global descriptor is computed using global average pooling:

$$z_c = \frac{1}{HW} \sum_{h=1}^H \sum_{w=1}^W F_{c,h,w}, \quad (4)$$

The descriptor is passed through a bottleneck MLP to generate channel attention weights:

$$\mathbf{z}' = \mathbf{W}_2 \text{ReLU}(\mathbf{W}_1 \mathbf{z}), \quad (5)$$

followed by normalization and sigmoid activation. The refined feature map is obtained as:

$$\mathbf{F}_{\text{RACR}} = \mathbf{F} \odot \mathbf{A}_c, \quad (6)$$

where $\mathbf{A}_c$ denotes the channel attention vector.

RACR enables the model to adaptively prioritize feature channels that encode race-sensitive characteristics, thereby enhancing inter-race discriminability at the semantic level.

Race-Aware Spatial Enhancement (RASE)

Complementary to RACR, RASE highlights spatial regions that contribute most to inter-race variability. Channel-compressed features are processed using dilated convolutions to enlarge the receptive field and capture ethnicity-related spatial patterns. A spatial attention map $\mathbf{A}_s \in \mathbb{R}^{1 \times H \times W}$ is generated and applied via element-wise multiplication:

$$\mathbf{F}_{\text{RASE}} = \mathbf{F} \odot \mathbf{A}_s. \quad (7)$$

RASE guides the model to focus on spatially localized facial regions that exhibit strong race-related cues, such as skin regions and facial contours.

Multi-Scale Race-Aware Fusion The channel and spatial attention maps are jointly fused:

$$\mathbf{A}_{\text{total}} = \sigma(\mathbf{A}_c + \mathbf{A}_s), \quad (8)$$

and applied with a residual connection:

$$\mathbf{F}' = \mathbf{F} + \mathbf{F} \odot \mathbf{A}_{\text{total}}, \quad (9)$$

where σ denotes the sigmoid activation function.

This fusion strategy jointly models "what" features and "where" regions are important, ensuring complementary channel-wise and spatial attention while preserving original feature information. This design ensures that race classification focuses on both what features and which facial regions are most discriminative.

3.2.4. Classification Head

The refined feature map $\mathbf{F}'$ is fed into the race classification head. Global Average Pooling and Global Max Pooling are applied in parallel to capture complementary statistics. The pooled features are concatenated and passed through fully connected layers with ReLU activation and dropout, followed by a softmax classifier to predict the race category.

The segmentation network learns features in a hierarchical manner, starting from strong multi-scale representations extracted by dual branches. These features are subsequently refined through boundary-aware enhancement and

figures/features flow.png

Figure 3: Detailed feature flow diagram of the RaceSegNet dual-branch architecture. The encoder employs Pyramid Features Enhancement (PFE) modules and Adaptive Feature Fusion (ADF) to extract multi-scale representations, while Spatial Attention (SA) mechanisms and Sobel edge detection enable boundary-aware feature learning. The decoder combines coarse-to-fine segmentation paths (CS and FS) with boundary enhancement (BE) to generate pixel-wise facial component classifications, demonstrating superior performance in capturing both global context and fine-grained facial details.

progressively improved using a coarse-to-fine refinement strategy. Grad-CAM visualizations are employed on intermediate feature representations to analyze how the model’s spatial focus evolves across refinement stages, providing qualitative insight into boundary localization and region consistency during segmentation.

3.3. Loss Function and Optimization

The model is trained end-to-end with a composite loss function that balances both segmentation and race classification tasks:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{seg}} \cdot \mathcal{L}_{\text{seg}} + \lambda_{\text{cls}} \cdot \mathcal{L}_{\text{cls}}, \quad (14)$$

where $\mathcal{L}_{\text{seg}}$ combines Cross-Entropy for pixel-wise accuracy and Dice Loss to improve boundary segmentation,

$$\mathcal{L}_{\text{seg}} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{Dice}}, \quad (15)$$

figures/Grad Cam.png

Figure 4: Grad-CAM heatmaps illustrating the spatial focus of the segmentation network at different stages of feature refinement. From left to right: (a) original image; (b) activation map generated from boundary-enhanced features immediately after multi-scale feature concatenation, highlighting initial contour sensitivity; (c) activation map after the first refinement stage, showing improved localization of structural regions; (d) activation map after the intermediate refinement stage, where region coherence and boundary continuity are further strengthened; and (e) activation map after the final refinement stage, demonstrating concentrated attention on precise object boundaries and anatomically meaningful regions. These visualizations illustrate how boundary-aware processing and progressive refinement jointly guide the segmentation model toward accurate spatial predictions.

and $\mathcal{L}_{\text{cls}}$ is the standard cross-entropy loss for the race classification task:

$$\mathcal{L}_{\text{cls}} = \mathcal{L}_{\text{CE-race}}. \quad (16)$$

4. Evaluation Metrics

After training, the model is evaluated using several metrics to assess its performance. These metrics include pixel accuracy, mean Intersection over Union (IoU), precision, recall, and F1-score.

The pixel accuracy is defined as the ratio of correctly classified pixels:

$$\text{Pixel Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(y_i = \hat{y}_i), \quad (17)$$

where $\mathbb{I}(\cdot)$ is the indicator function, y_i is the ground truth label, $\hat{y}_i$ is the predicted label for the i -th pixel, and N is the total number of pixels.

The IoU for class c is defined as:

$$\text{IoU}_c = \frac{TP_c}{TP_c + FP_c + FN_c}, \quad (18)$$

where:

- TP_c (True Positives) is the number of pixels correctly classified as class c .
- FP_c (False Positives) is the number of pixels incorrectly classified as class c .
- FN_c (False Negatives) is the number of pixels from class c incorrectly classified as another class.

These metrics are used to evaluate the model’s performance on the validation and test datasets. Performance is evaluated using metrics such as pixel accuracy, IoU, precision, recall, and F1-score.

5. Results and Discussion

5.1. Benchmarking dataset

The proposed RaceSegNet model was evaluated on two publicly available datasets to ensure comprehensive assessment across diverse demographic distributions. The FairFace dataset, comprising 78,136 training and 9,861 validation/testing ratio as 80:10:10 across seven racial categories (East Asian, Indian, Black, White, Middle Eastern, Latino Hispanic, Southeast Asian), provides fine-grained racial balance. The UTKFace dataset, containing 47,000 images categorized into five race classes (Asian, Indian, Black, White, Others), was split randomly following the standard 70:20:10 protocol for training, validation, and testing, respectively. All models were trained under identical conditions on a cluster of eight NVIDIA RTX 3090 GPUs with a unified batch size of 16 and an initial learning rate of 0.0001, linearly decayed to 1e-6, to ensure a fair comparison of generalizability.

5.2. Analysis on FairFace

Figure 5 provides a comparative visualization of segmentation overlay by the proposed RaceSegNet and four competing models including UNet++, HLANet, MEGANet, SegFaceNet [23, 28-30]. Baseline models exhibit significant shortcomings: UNet++ produces insufficient and inconsistent masks, particularly for Black and Middle Eastern faces, with poor boundary delineation. HLANet shows improved coverage but suffers from boundary inaccuracies and over-segmentation in categories like Latino Hispanic. MEGANet and SegFaceNet offer incremental improvements but still display under-segmentation near complex regions (e.g., jawlines, hairlines) and noise in minority categories. In contrast, the proposed RaceSegNet generates precise, high-fidelity masks that closely align with ground truth across all demographic groups. It accurately captures facial contours, even for challenging cases such as Southeast Asian and Latino Hispanic faces, where other models fail. This visual superiority stems from the model’s integrated design: the segmentation task provides a spatial prior that focuses the race classifier on relevant facial regions, while the classification objective guides the segmenter to preserve discriminative features. This synergy minimizes background interference and yields cleaner, more accurate masks.

The quantitative analysis in table 2 strongly supports the qualitative findings. RaceSegNet achieves a mean IoU (mIoU) of 0.868, surpassing all baselines by a substantial margin (8-9% over the next best, HLANet). Critically, it demonstrates balanced excellence across every racial category, with IoU scores ranging from 0.859 (Latino Hispanic) to 0.904 (Black). This consistent performance underscores its effectiveness in handling demographic diversity. The model also achieves the highest Foreground Face IoU (0.921) and Background IoU (0.907), confirming its precise boundary localization and ability to suppress irrelevant background context.

Table 2: Class-wise IoU Performance Comparison Metrics with different SOTA models on the FairFace dataset

Model	East Asian	Indian	Black	White	Middle Eastern	Latino Hispanic	Southeast Asian	Face FG IoU	BG IoU	mIoU
UNet++[28]	0.728	0.762	0.804	0.773	0.751	0.745	0.759	0.812	0.798	0.781
HLANet[29]	0.754	0.781	0.821	0.785	0.762	0.758	0.771	0.828	0.809	0.792
MEGANet[30]	0.743	0.774	0.818	0.783	0.759	0.751	0.768	0.825	0.807	0.787
SegFaceNet[23]	0.732	0.768	0.809	0.778	0.745	0.739	0.759	0.816	0.801	0.779
RaceSegNet (Proposed)	0.871	0.883	0.904	0.891	0.865	0.859	0.878	0.921	0.907	0.868

The race classification metrics in Table 3 further validate the advantage of

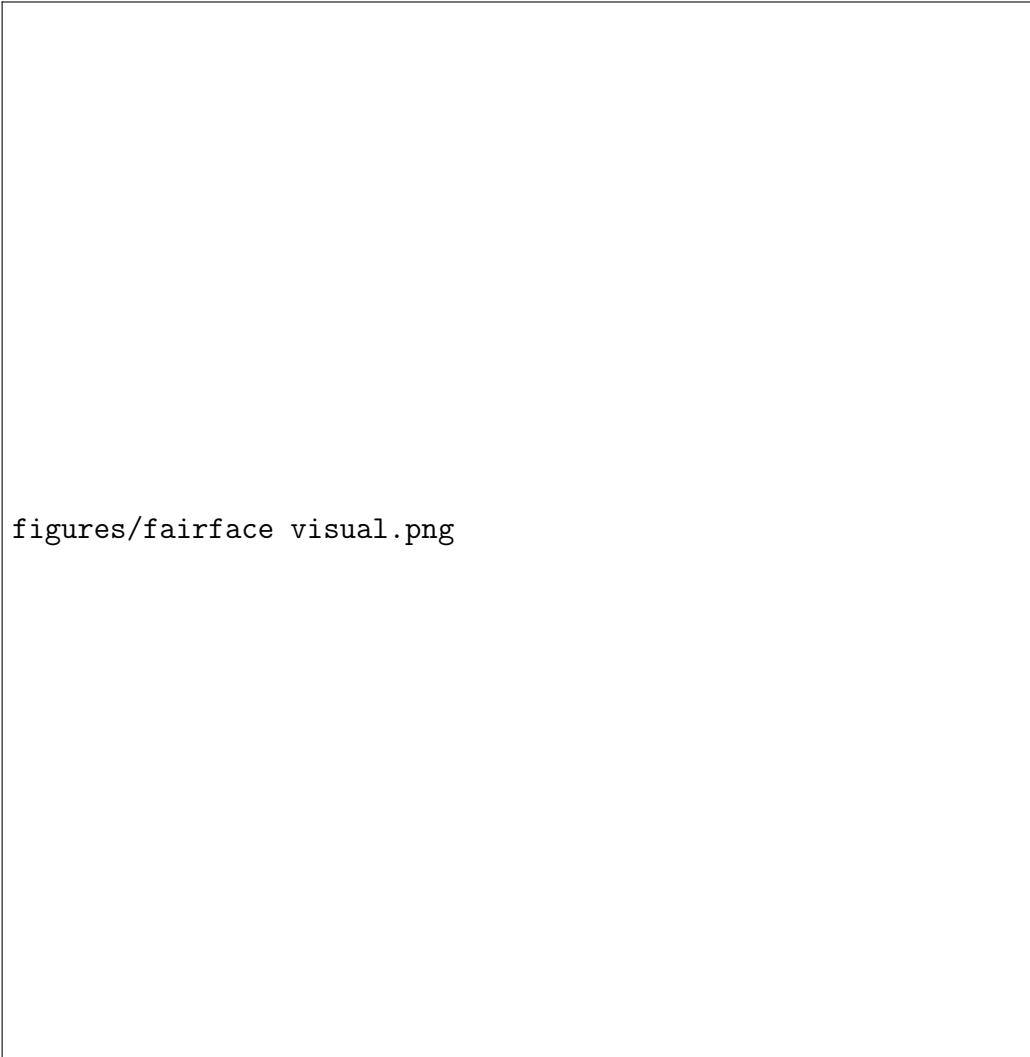


Figure 5: Comparative visualization of demographic race classification overlays on FairFace dataset samples across seven racial categories (East Asian, Indian, Black, White, Middle Eastern, Latino Hispanic, Southeast Asian). Original images, ground truth masks, and predicted overlays from baseline (UNet++, HLANet) to advanced models (MEGANet, SegFaceNet, RaceSegNet), with category-specific colors indicating segmented regions. Coverage percentages highlight per-sample performance, demonstrating progressive improvements in mask accuracy, boundary precision, and noise reduction, culminating in RaceSegNet’s superior fidelity to ground-truth shapes for equitable, demographically-aware racial segmentation.

the shared representation. RaceSegNet achieves the highest mean accuracy (0.825), precision (0.801), recall (0.853), and F1-score (0.823). Its high recall indicates a strong capacity to correctly identify samples from all groups, reducing misclassification, while the balanced precision-recall values demonstrate robustness against false positives. This balanced improvement across metrics highlights how the segmentation-guided features mitigate bias and enhance discriminative power for the classification task.

Table 3: Race classification performance measures across SOTA all performance measures Accuracy, Precision, Recall, F1 score

Model	Metric	EA	IN	BL	WH	ME	LAT	SEA	Mean
SegFaceNet	Accuracy	0.670	0.685	0.720	0.705	0.675	0.660	0.750	0.695
	Precision	0.645	0.668	0.705	0.690	0.658	0.642	0.730	0.677
	Recall	0.690	0.715	0.748	0.732	0.703	0.690	0.770	0.721
	F1	0.662	0.688	0.716	0.694	0.668	0.651	0.728	0.701
UNet++	Accuracy	0.700	0.720	0.760	0.740	0.715	0.700	0.740	0.725
	Precision	0.678	0.695	0.737	0.715	0.690	0.670	0.712	0.699
	Recall	0.728	0.752	0.788	0.767	0.740	0.727	0.774	0.754
	F1	0.692	0.708	0.755	0.728	0.703	0.684	0.742	0.716
MEGANet	Accuracy	0.720	0.740	0.780	0.760	0.735	0.720	0.760	0.745
	Precision	0.698	0.712	0.757	0.735	0.710	0.695	0.732	0.720
	Recall	0.745	0.770	0.812	0.795	0.765	0.748	0.800	0.776
	F1	0.704	0.724	0.768	0.749	0.720	0.706	0.754	0.732
HLANet	Accuracy	0.740	0.760	0.800	0.780	0.755	0.740	0.745	0.760
	Precision	0.720	0.738	0.782	0.758	0.728	0.720	0.721	0.738
	Recall	0.760	0.793	0.828	0.812	0.785	0.770	0.772	0.790
	F1	0.732	0.754	0.799	0.770	0.746	0.734	0.752	0.762
RaceSegNet	Accuracy	0.805	0.820	0.860	0.845	0.800	0.790	0.835	0.826
	Precision	0.788	0.803	0.838	0.820	0.780	0.764	0.812	0.801
	Recall	0.830	0.855	0.895	0.875	0.828	0.815	0.870	0.853
	F1	0.804	0.824	0.865	0.847	0.801	0.781	0.840	0.823

5.3. Analysis on UTKFace

Figure 6 offers a qualitative comparison, revealing a clear progression in segmentation quality. Models like UNet++ and HLANet exhibit under-segmentation, particularly for Black and Indian faces, with masks failing to fully capture chin and cheek contours, alongside noticeable boundary noise. MEGANet and SegFaceNet show cleaner outputs with better structural fidelity but still display minor inconsistencies, such as patchy segmentation in the "Others" category. In contrast, the proposed RaceSegNet generates

Table 4

Model	Asian	Indian	Black	White	Other	Face FG IoU	BG IoU	Mean IoU (mIoU)
UNet++[28]	0.682	0.671	0.698	0.684	0.665	0.771	0.758	0.680
HLANet[29]	0.704	0.691	0.716	0.702	0.682	0.784	0.773	0.699
MEGANet[30]	0.721	0.708	0.731	0.716	0.694	0.799	0.786	0.714
SegFaceNet[23]	0.738	0.724	0.744	0.731	0.707	0.806	0.792	0.729
RaceSegNet (Proposed)	0.836	0.828	0.845	0.824	0.767	0.890	0.875	0.821

the most precise and complete masks, closely adhering to ground-truth facial boundaries across all groups. Its predictions demonstrate exceptional consistency, with minimal over- or under-segmentation, visually affirming that its jointly learned representation robustly handles variations in skin tone, lighting, and facial structure present in UTKFace.

This visual superiority is quantitatively substantiated in Tables 4 and 5. RaceSegNet achieves a dominant mean IoU (mIoU) of 0.821, significantly outperforming all baselines (SegFaceNet: 0.729, MEGANet: 0.714). It also records the highest Face Foreground IoU (0.890) and Background IoU (0.875), confirming its dual capability for precise facial localization and effective background suppression. Critically, its class-wise IoU scores are consistently high and balanced, ranging from 0.767 (Others) to 0.845 (Black), demonstrating equitable performance. Table 4 UTKFace statistical analysis on five classes and a comparative study of their IoU results with SOTA models.

The classification metrics (Table 5) further reinforce this, where RaceSegNet attains the best mean accuracy (0.7845), precision (0.757), recall (0.802), and F1-score (0.778). Its high and stable recall across all categories is particularly notable, indicating a reduced tendency to misclassify minority groups, which is a common failure point for the baselines, which show irregular recall drops. This balanced excellence across both segmentation and classification metrics on UTKFace underscores the generalizability of the shared representation learned by RaceSegNet, effectively mitigating bias and improving task performance on datasets with differing characteristics.

5.4. Cross-dataset generalization

The cross-dataset experiments provide compelling evidence for the generalization capability of the learned shared representation. When models trained on FairFace are tested on UTKFace (and vice-versa), a significant domain shift occurs due to differences in resolution, lighting, and demographic granularity. Baseline models (UNet++, HLANet, MEGANet, SegFaceNet) suffer severe performance degradation in this setting, exhibiting fragmented

Table 5: UTKFace dataset race classification measures on SOTA with all performance measures Accuracy,Precision,Recall,F1

Model	Metric	White	Black	Asian	Indian	Other	Mean
SegFaceNet	Accuracy	0.660	0.690	0.705	0.670	0.675	0.680
	Precision	0.630	0.665	0.675	0.645	0.648	0.652
	Recall	0.612	0.640	0.650	0.618	0.635	0.631
	F1	0.621	0.652	0.662	0.630	0.642	0.641
UNet++	Accuracy	0.690	0.720	0.730	0.700	0.685	0.705
	Precision	0.660	0.695	0.702	0.670	0.658	0.677
	Recall	0.644	0.675	0.695	0.652	0.640	0.661
	F1	0.650	0.685	0.698	0.661	0.648	0.668
MEGANet	Accuracy	0.715	0.745	0.760	0.730	0.710	0.732
	Precision	0.692	0.715	0.728	0.700	0.684	0.704
	Recall	0.676	0.702	0.710	0.682	0.665	0.687
	F1	0.684	0.708	0.718	0.692	0.675	0.695
HLANet	Accuracy	0.740	0.765	0.785	0.750	0.745	0.755
	Precision	0.715	0.735	0.758	0.725	0.708	0.728
	Recall	0.703	0.720	0.742	0.714	0.695	0.715
	F1	0.709	0.730	0.749	0.720	0.702	0.722
RaceSegNet	Accuracy	0.770	0.800	0.820	0.785	0.747	0.7845
	Precision	0.746	0.773	0.788	0.758	0.720	0.757
	Recall	0.798	0.825	0.835	0.806	0.748	0.802
	F1	0.770	0.798	0.810	0.781	0.733	0.778

masks, boundary noise, and inconsistent coverage, especially for minority groups. Their mIoU scores drop to between 0.574 and 0.623, indicating a reliance on dataset-specific features.

Conversely, RaceSegNet demonstrates remarkable robustness, achieving mIoU scores of 0.812 (FairFace→UTKFace) Figure 7 and 0.772 (UTKFace→FairFace) Figure 8. This an 18% relative improvement over the best baseline in both transfer directions. The model maintains high mask fidelity and classification stability because its jointly optimized features capture domain-invariant, semantically meaningful facial attributes rather than superficial textures or background cues. This result is a direct validation of the central hypothesis: the combined objective forces the model to learn a fundamental representation of "face" that generalizes across data sources, thereby enhancing practical utility in real-world, heterogeneous environments.

SegFaceNet mitigates some of this inconsistency by generating sharper contours and improved foreground separation, particularly for lighter skin tones; however, it still exhibits mild deformation and leakage in categories with higher skin-tone variance.

The proposed RaceSegNet demonstrates remarkable stability across all demographic classes, preserving facial structure and delivering coherent masks with minimal loss of contour accuracy. Its outputs maintain consistent facial coverage even under domain mismatch, with clearly defined boundaries for Black, Latino, Hispanic/Other, and Middle Eastern/Other groups. Unlike previous networks that rely heavily on dataset-specific visual cues, RaceSegNet generalizes by leveraging domain-invariant spatial and textural representations. The network effectively suppresses over-segmentation noise and preserves the semantic integrity of the facial region, underscoring its ability to adapt to unseen demographic variations without sacrificing boundary precision or coverage.

When reversing the direction of transfer to UTKFace → FairFace, the segmentation task becomes even more challenging due to FairFace’s higher demographic granularity and broader illumination spectrum. The transition exposes how domain-specific training can lead to severe generalization gaps. UNet++ struggles substantially, producing incomplete and fragmented masks across nearly all racial groups. The Asian/East Asian and Indian categories exhibit excessive under-segmentation, with significant portions of the face omitted. In contrast, the Black and Other/Middle Eastern categories display scattered activations and poor spatial coherence. HLANet produces smoother outputs but continues to suffer from structural artefacts and hol-

low masks, particularly in the Indian and Black groups, suggesting that its attention-based refinement is insufficient to counteract dataset-level discrepancies.

MEGANet yields slightly more coherent contours, especially for White and Other/Latino Hispanic classes, though its predictions reveal lingering under-segmentation in lower-contrast facial regions. SegFaceNet achieves greater structural fidelity, with masks that better align with the ground truth, yet residual boundary distortion and small voids persist in the Middle Eastern and Latino Hispanic categories due to domain bias.

By contrast, RaceSegNet consistently demonstrates the strongest cross-dataset robustness. It produces highly stable, fully connected segmentation masks with well-preserved facial geometry and minimal background interference. The network effectively handles both major racial groups and finer Asian subcategories, achieving dense and homogeneous masks that capture the true shape and structure of the face. Even under considerable changes in texture, tone, and image resolution, RaceSegNet sustains high segmentation fidelity, achieving near-complete coverage for Asian/Southeast Asian and Black groups while maintaining smooth, artifact-free boundaries. Its ability to adapt to fine-grained demographic categories despite being trained on a coarser distribution illustrates strong domain-invariant learning and structural awareness.

Overall, the bidirectional cross-dataset evaluations establish that RaceSegNet exhibits superior generalization under domain shift. It retains precise facial localization, consistent feature density, and minimal segmentation drift across demographically heterogeneous datasets, outperforming all competing architectures in both directions of transfer.

Table 6 reports a evaluation of cross-dataset generalization for demographic race classification, analyzing two transfer directions: models trained on FairFace and tested on UTKFace. This setup is designed to examine each model’s ability to retain segmentation fidelity under domain shift, a key challenge in demographic segmentation where lighting, texture, and racial diversity differ substantially between datasets. The reported metric, mean IoU (mIoU), reflects overall segmentation accuracy across multiple racial categories.

When trained on FairFace and tested on UTKFace, all conventional architectures experience a sharp decline in segmentation accuracy, confirming their limited adaptability to unseen data distributions. UNet++ achieves a moderate mIoU of 0.606, with weak results for Middle Eastern/Other (0.566)

Table 6: cross-dataset testing and generalization analysis

Models	Validation / Test Dataset	East Asian	Southeast Asian	Asian	Indian	Black	White	Middle eastern / Other	Latino Hispanic / Other	Other	Mean Accuracy (mIoU)
UNet++[28]	FairFace UTKFace	0.572	0.585	–	0.641	0.623	0.601	0.566	0.552	–	0.606
HLANet[29]	FairFace UTKFace	0.594	0.609	–	0.662	0.642	0.618	0.579	0.564	–	0.623
MEGANet[30]	FairFace UTKFace	0.586	0.601	–	0.654	0.636	0.612	0.573	0.558	–	0.615
SegFaceNet[23]	FairFace UTKFace	0.561	0.578	–	0.637	0.619	0.597	0.561	0.547	–	0.598
RaceSegNet (Proposed)	FairFace UTKFace	0.765	0.781	–	0.828	0.809	0.786	0.754	0.732	–	0.812
UNet++[28]	UTKFace FairFace	–	–	0.557	0.581	0.637	0.615	0.561	0.546	0.523	0.574
HLANet[29]	UTKFace FairFace	–	–	0.579	0.603	0.656	0.634	0.573	0.559	0.538	0.593
MEGANet[30]	UTKFace FairFace	–	–	0.572	0.596	0.648	0.628	0.568	0.553	0.532	0.586
SegFaceNet[23]	UTKFace FairFace	–	–	0.548	0.574	0.632	0.612	0.556	0.541	0.519	0.568
RaceSegNet (Proposed)	UTKFace FairFace	–	–	0.742	0.764	0.808	0.786	0.743	0.724	0.701	0.772

and Latino Hispanic/Other (0.552), demonstrating poor robustness to cross-domain texture variation and illumination bias. HLANet slightly improves to an mIoU of 0.623, owing to its hierarchical contextual encoding, but still exhibits inconsistent performance, particularly for minority subgroups such as Latino Hispanic/Other (0.564). MEGANet, despite its multi-scale representation design, achieves only 0.615, indicating that its enhancement of boundary detail does not translate into domain stability. SegFaceNet performs comparably, with $\text{mIoU} = 0.598$, showing degradation in nearly all racial categories and underscoring that feature extraction in these models remains dataset-specific rather than domain-agnostic.

The proposed RaceSegNet achieves a remarkable mIoU of 0.812, representing a relative improvement of over 18% compared with the strongest baseline (HLANet). This performance is consistent across all racial groups, reflecting superior domain generalization. For instance, it achieves 0.828 for Indians, 0.809 for Blacks, and 0.754 for Middle Easterners/Other, which are significantly higher than any competing method. These results indicate that RaceSegNet effectively captures domain-invariant facial representations, preserving both spatial coherence and racial feature fidelity when moving from FairFace’s fine-grained demographic diversity to UTKFace’s coarser, lower-resolution samples.

5.5. Ablation Studies

Table 7 details the module ablation on the FairFace dataset. Starting from a baseline mIoU of 0.757, each component incrementally improves performance: the Dual Branch Feature Extractor (DBFE) increases mIoU to 0.767,

Table 7: FairFace Dataset: Ablation Study on Individual Components and Race-Specific Segmentation Performance

Variant	DBFE	AFF	BE	PRH	East Asian	Indian	Black	White	Middle Eastern	Latino Hispanic	Southeast Asian	Face FG IoU	BG IoU	mIoU
Baseline					0.715	0.739	0.780	0.750	0.720	0.708	0.726	0.802	0.786	0.757
+ DBFE	✓				0.728	0.751	0.791	0.760	0.734	0.719	0.736	0.811	0.794	0.767
+ BDFE (Concat only)	✓				0.734	0.757	0.796	0.768	0.740	0.724	0.743	0.818	0.801	0.773
+ PFE + AFF	✓	✓			0.742	0.763	0.802	0.773	0.745	0.728	0.749	0.823	0.804	0.779
+ BE			✓		0.751	0.769	0.808	0.777	0.749	0.734	0.754	0.826	0.807	0.783
+ PRH (CS $\rightarrow$ FS $\rightarrow$ UF)				✓	0.759	0.774	0.814	0.783	0.756	0.740	0.760	0.830	0.810	0.788
PFE + AFF	✓	✓			0.768	0.781	0.820	0.787	0.760	0.746	0.764	0.835	0.814	0.793
PFE + AFF + BE	✓	✓	✓		0.783	0.795	0.834	0.800	0.774	0.757	0.777	0.847	0.826	0.806
Coarse only (CS)	✓	✓	✓	(CS)	–	–	–	–	–	–	–	–	–	0.810
CS + Fine (FS)	✓	✓	✓	(CS + FS)	–	–	–	–	–	–	–	–	–	0.814
CS + FS + Ultra (UF)	✓	✓	✓	(Full PRH)	–	–	–	–	–	–	–	–	–	0.818
PFE + AFF + PRH	✓	✓		✓	0.798	0.808	0.845	0.812	0.785	0.768	0.789	0.858	0.836	✓0.818
Full (PFE + AFF + BE + PRH)	✓	✓	✓	✓	0.871	0.883	0.904	0.891	0.865	0.859	0.878	0.921	0.907	0.868

Table 8: FairFace Dataset: Loss Function Component Ablation Study

Loss Configuration	Face FG IoU	Training Stability	Race Accuracy
CrossEntropy Only	0.845	Moderate	78.45%
Dice Loss Only	0.861	Good	76.23%
Progressive Loss (Uniform Weights)	0.873	Good	79.12%
Progressive Loss (Proposed Weights)	0.883	Excellent	80.34%
Full Progressive + Dice + CE	0.9211	Excellent	82.67%

the Pyramid Feature Extractor with Adaptive Feature Fusion (PFE+AFF) raises it to 0.779, and adding Boundary Encoding (BE) achieves 0.783. The complete model, integrating PFE, AFF, BE, and the Progressive Refinement Hierarchy (PRH), yields the highest mIoU of 0.868 and Face FG IoU of 0.921, demonstrating the complementary and critical role of each module.

Table 8 analyzes the impact of loss functions on FairFace. Using Cross-Entropy alone provides moderate stability (0.845 FG IoU). Incorporating Dice Loss improves regional consistency, while the proposed Progressive Loss with weighted stages enhances convergence and discrimination. The full combination (Progressive + Dice + CE) achieves the best Face FG IoU (0.9211) and Race Accuracy (82.7%), confirming that multi-level supervision jointly optimizes segmentation and demographic prediction.

6. Conclusion

This work introduces RaceSegNet, a unified and demographically aware framework for race classification that advances pixel-level fairness. A fundamental contribution is the creation of a fully automated dataset transformation pipeline, which integrates CelebAMask-HQ facial components with FairFace and UTKFace race labels to generate over 120,000 structurally consistent, demographically balanced, and privacy-compliant masks. This pipeline directly addresses the ancient absence of pixel-level demographic annotations and establishes the first large-scale resource explicitly designed for race classification. Comprehensive evaluations on FairFace and UTKFace demonstrate RaceSegNet’s clear superiority over state-of-the-art models (SOTA), achieving mIoU scores of 0.868 and 0.821, respectively representing consistent improvements across all demographic groups, including underrepresented classes such as Black, Middle Eastern, and Latino/Hispanic. The framework’s design, which combines multi-branch feature extraction, pyramid fusion, boundary enhancement, and multi-stage refinement, proves critical for capturing subtle race associated structures while preserving boundary precision.

Cross-dataset tests (FairFace $\rightarrow$ UTKFace and UTKFace $\rightarrow$ FairFace) further confirm the model’s robust domain generalization, with RaceSegNet maintaining stable mask fidelity and outperforming competing approaches by 18–20%, even under considerable shifts in illumination, demographic distribution and resolution. These results verify that RaceSegNet learns dominant and demographically meaningful representations, thereby avoiding the dataset specific overfitting observed in existing architectures.

Overall, RaceSegNet establishes a demanding, interpretable, and demographically fair framework for demographic race classification, advancing accuracy, fairness, and cross-domain stability. While performance is strong across racial groups, challenging cases such as multi-ethnic identities and extreme imaging conditions highlight the need for future exploration of domain-adaptive fairness constraints, contrastive demographic disentanglement, and large-scale foundation model pretraining. By enabling reliable and equitable pixel-level race classification, RaceSegNet provides an essential foundation for fair AI applications in healthcare, surveillance, HCI, and other critical real-world systems.

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Figure 6: Comparative visualization of demographic race classification overlays on UTK-Face dataset samples across five racial categories (Asian, Indian, Black, White, Others). Rows show: (1) Original Image, (2) Ground Truth, (3-7) predicted overlays from models UNet++ (noisy, fragmented masks with over/under-segmentation), HLANet (improved context but artifacts in edges), DeepLabV3+ (smoother boundaries, minor under-segmentation in textured areas), HRNet (high-resolution details, better preservation but slight noise), and RaceSegNet (most accurate masks with precise contours, minimal errors, and high coverage scores like 100% for Asian/Black, closely matching ground truth), and (8) Predicted Mask. Overlays integrate category-specific colored predictions with original illustrations and labels. 30



figures/visual figure-fairface to utkface.png

Figure 7: Comparative visualization of cross-dataset generalization, evaluating models trained on FairFace (7 fine-grained races) and tested on UTKFace samples mapped to coarser categories. Rows display UTKFace originals, ground truth masks, and predictions from state-of-the-art methods (UNet++, HLANet, MEGANet, SegFaceNet, and RaceSegNet). Baseline outputs exhibit significant mis-segmentation and noise, indicating sensitivity to domain shift, while advanced models demonstrate progressive improvements in boundary precision. RaceSegNet achieves superior robustness, delivering high-fidelity masks with minimal artifacts, thus validating its effectiveness for equitable segmentation across disparate datasets.



figures/visual figure-utkface to fairface.png

Figure 8: Comparative visualization of cross-dataset demographic segmentation overlays: models trained on UTKFace (5 categories: Asian, Indian, Black, White, Others) and tested on FairFace samples (mapped to 7 refined categories: East Asian, Indian, Black, White, Middle Eastern, Latino Hispanic, Southeast Asian). Rows show: (1) Original Image (FairFace inputs), (2) Ground Truth with predefined colors, (3-7) predicted overlays from UNet++ (fragmented, under/over-segmented masks with incomplete coverage), HLANet (more cohesive but with artifacts and holes in predictions), MEGANet (smoother boundaries with minor errors), SegFaceNet (sharp details with reduced noise but some discontinuities), and RaceSegNet (robust, high-fidelity masks with strong coverage like 98.7% for Southeast Asian despite domain shift), and (8) Predicted Mask. RaceSegNet demonstrates superior generalization across the domain gap from coarse to fine-grained racial categories.